

Computer Networks Lab Fall 2025

Week 14

VLANs and ACL

Configuration of Virtual LANs

1. Introduction:

1.1. Virtual LANs:

A VLAN is a group of devices on one or more LANs that are configured to communicate as if they were attached to the same wire, when in fact they are located on a number of different LAN segments. Because VLANs are based on logical instead of physical connections, they are extremely flexible.

VLANs define broadcast domains in a Layer 2 network. A broadcast domain is the set of all devices that will receive broadcast frames originating from any device within the set. Broadcast domains are typically bounded by routers because routers do not forward broadcast frames. Layer 2 switches create broadcast domains based on the configuration of the switch. Switches are multiport bridges that allow you to create multiple broadcast domains. Each broadcast domain is like a distinct virtual bridge within a switch.

You can define one or many virtual bridges within a switch. Each virtual bridge you create in the switch defines a new broadcast domain (VLAN). Traffic cannot pass directly to another VLAN (between broadcast domains) within the switch or between two switches. To interconnect two different VLANs, you must use routers or Layer 3 switches.

1.2. VLAN parameters:

You can set the following parameters when you create a VLAN in the management domain:

- VLAN number
- VLAN name
- VLAN state (active or suspended)



National University of Computer and Emerging Sciences (NUCES) Islamabad

1.3. Trunk:

A trunk is a communications line or link designed to carry multiple [signals](#) simultaneously to provide network access between two points. Trunks typically connect switching centers in a communications system. The signals can convey any type of communications data. A trunk can consist of multiple wires, cables or [fiber optic](#) strands bundled together to maximize the available bandwidth in a single physical cable, or it can consist of a single high-capacity link over which many signals are [multiplexed](#). Data networks use two types of trunks. First, trunks can carry data from multiple local area networks ([LANs](#)) or virtual LANs ([VLANs](#)) across a single interconnect between switches or routers, called a trunk port. Second, trunks can bond or aggregate multiple physical links to create a single, higher-capacity, more reliable logical link, which is called port trunk.

4. Walk through task:

4.1. Task1:

Construct a topology that have two switches attached with each other with copper cross through wire and four PC's attached with them, a pair with each switch, configure Virtual LAN in this configuration and show the results.

To configure this, we have to execute following steps:

1. Open cisco packet tracer.
2. Add two 2690 switches in the workspace.
3. Add four generic PC's in the workspace.
4. Connect two switches with each other using copper cross over wire with fa 0/24 port.
5. Connect each pair of generic computers with switch using copper straight through wire.
6. The topology should be seen like this:

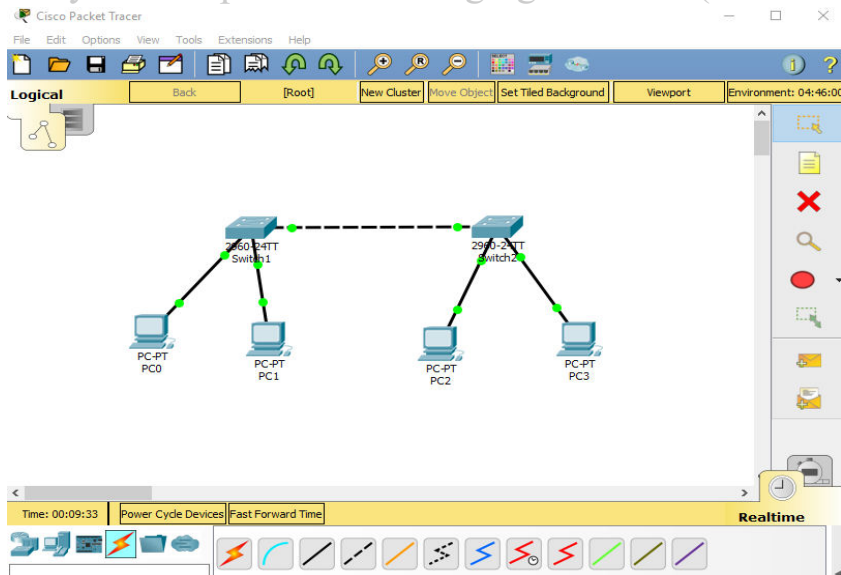


Figure 1: topology connection

7. Now add notes on the workspace in the front of every network entity with IP address and also mention the name of virtual LAN.

8. The workspace should be seen like this:

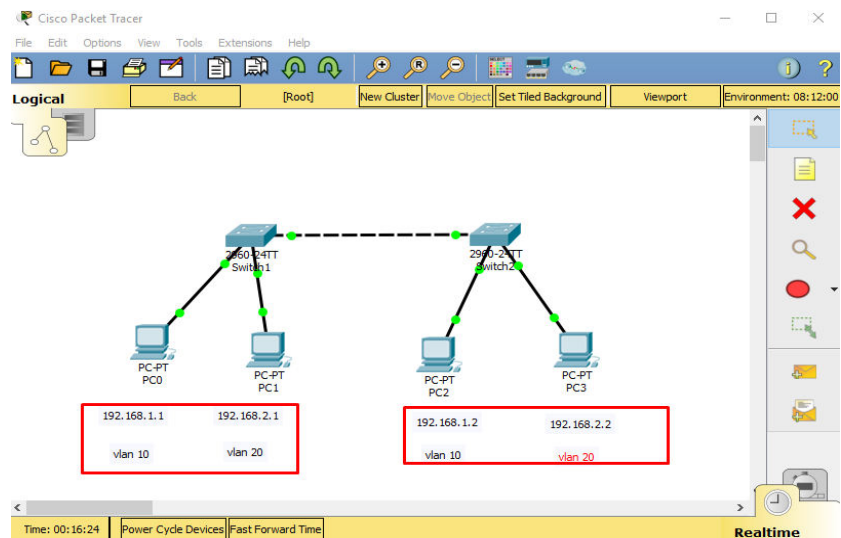


Figure 2: captions of the network entities.

9. Click on the switch 1 and go in command line interface
10. Go in the privileged mode by using command **enable**.
11. Go in the configuration mode by using command **configure terminal**.
12. Now to give proper name to virtual LAN's, write command **vlan "mention port number"** and press enter.
13. Write command **name "give name of the virtual Lan"**.

14. Write command **exit** and press enter.
15. To give name to other Vlan, repeat steps 12,13 and 14.
16. The commands in the switch CLI can be seen the figure below:

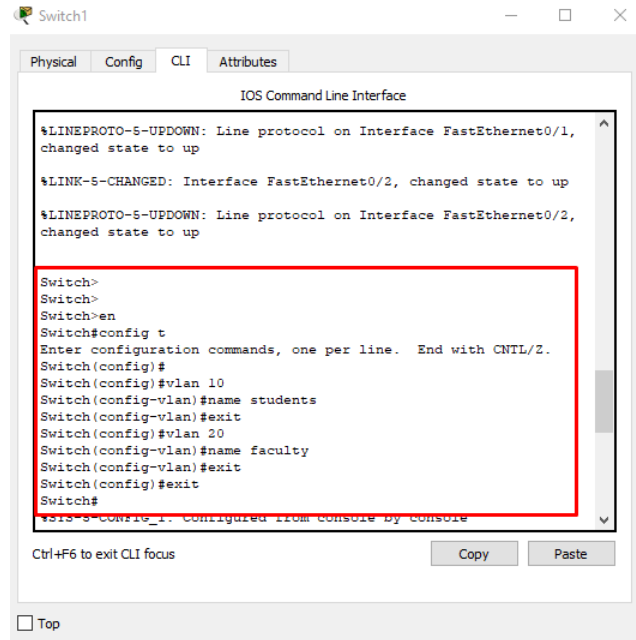


Figure 3: CLI of the switch.

17. To verify that the name has been given to the Virtual LAN's, write command **show vlan brief**.
18. The results will be like this:

```
Switch0
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan vrief
^
% Invalid input detected at '^' marker.

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   students                active
20   faculty                 active
1002 fddi-default           active
1003 token-ring-default     active
1004 fddinet-default         active
1005 trnet-default           active
```

Figure 4: results of verification of the vlan names

19. Now again come in the configuration mode and for the configuration of Vlan on the switch 1, go in the configuration of interface fa 0/1 and write command **switchport access vlan 10** and press enter and write command **exit** and press enter.
20. In the same way go in the configuration of fa 0/2 and write command **switchport access vlan 20** and press enter and write command **exit** and press enter.
21. The command line interface of the switch 1 should be like this:

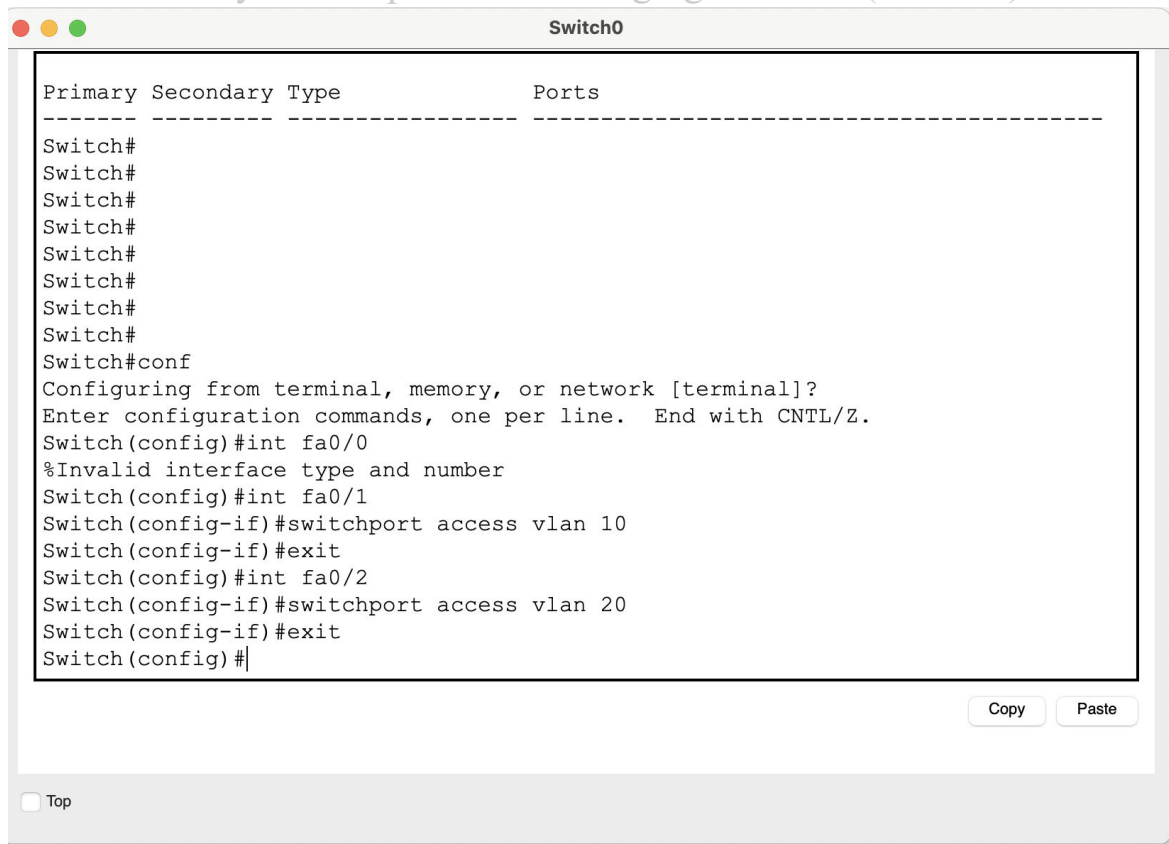


Figure 5: shows the command line interface of switch 1

22. For verification of vlan configuration write command do show vlan. You should receive these results:

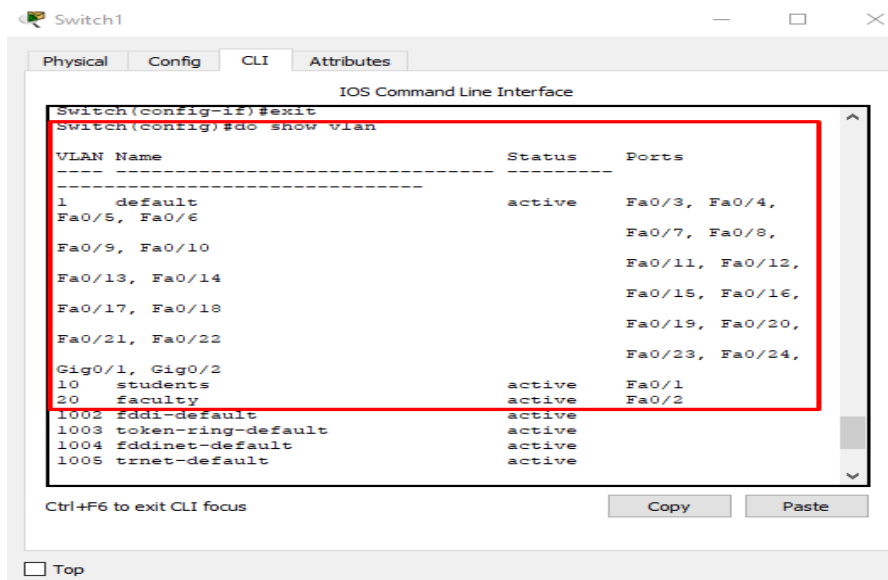
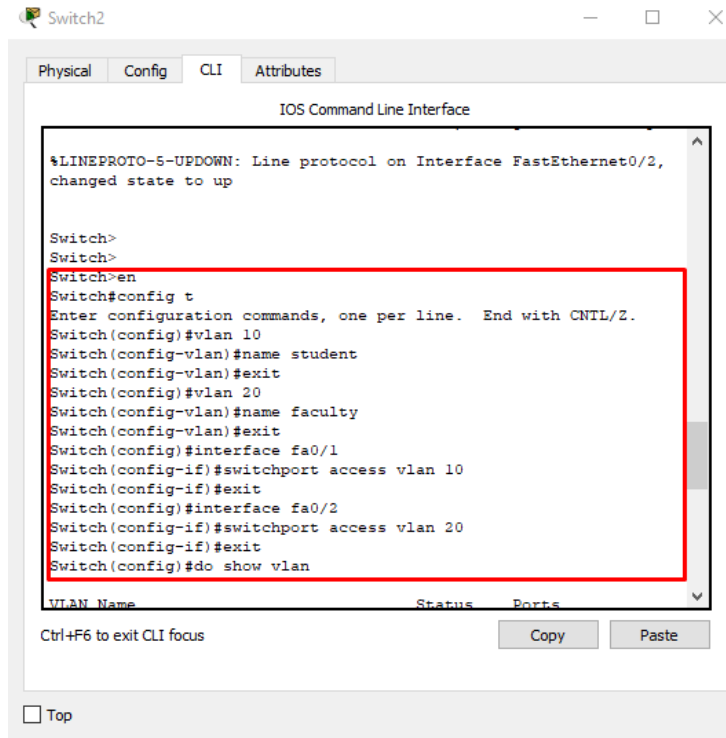


Figure 6: verification of vlan configuration

1. Now repeat all the steps as we have done for switch 1 for the switch 2 including name setting and vlan

National University of Computer and Emerging Sciences (NUCES) Islamabad
configuration in same steps, the command line interface for the switch2 is attached below:



```

Switch2
Physical Config CLI Attributes
IOS Command Line Interface

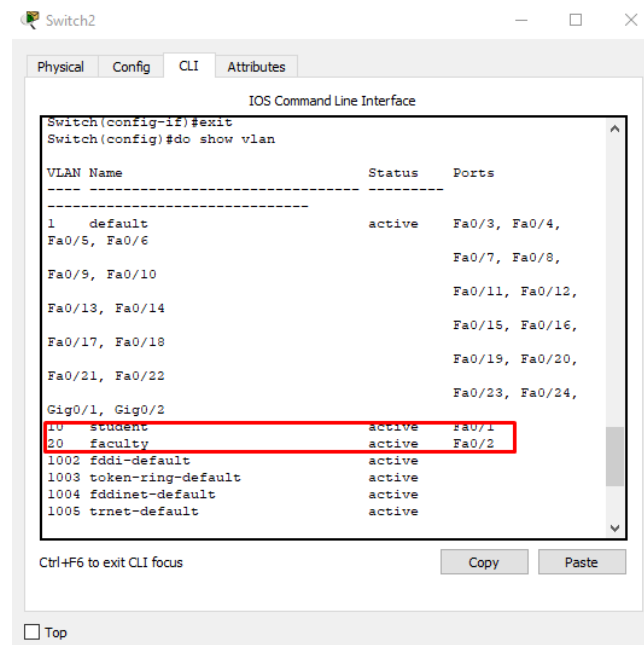
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

Switch>
Switch>
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name student
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name faculty
Switch(config-vlan)#exit
Switch(config)#interface fa0/1
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fa0/2
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#do show vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/3, Fa0/4,
Fa0/5, Fa0/6
Fa0/7, Fa0/8,
Fa0/9, Fa0/10
Fa0/13, Fa0/14
Fa0/17, Fa0/18
Fa0/21, Fa0/22
Fa0/23, Fa0/24,
Gig0/1, Gig0/2
10   student                active    Fa0/1
20   faculty                active    Fa0/2
1002 fddi-default          active
1003 token-ring-default  active
1004 fddinet-default      active
1005 trnet-default        active
  
```

Figure 7: configuration of Switch 2

2. The results are given below:



```

Switch2
Physical Config CLI Attributes
IOS Command Line Interface

Switch(config-if)#exit
Switch(config)#do show vlan

VLAN Name                Status    Ports
-----
1    default                active    Fa0/3, Fa0/4,
Fa0/5, Fa0/6
Fa0/7, Fa0/8,
Fa0/9, Fa0/10
Fa0/13, Fa0/14
Fa0/17, Fa0/18
Fa0/21, Fa0/22
Fa0/23, Fa0/24,
Gig0/1, Gig0/2
10   student                active    Fa0/1
20   faculty                active    Fa0/2
1002 fddi-default          active
1003 token-ring-default  active
1004 fddinet-default      active
1005 trnet-default        active
  
```

Figure 8: shows the results of switch2 for vlan

National University of Computer and Emerging Sciences (NUCES) Islamabad

3. After this configure trunk mode in switch 1 by going in the configuration mode of switch 1 and write **command interface fa 0/24** and press enter.
4. For enabling of trunk mode in switch 1 write command **switchport mode trunk** and press enter and write command **exit** and press enter.

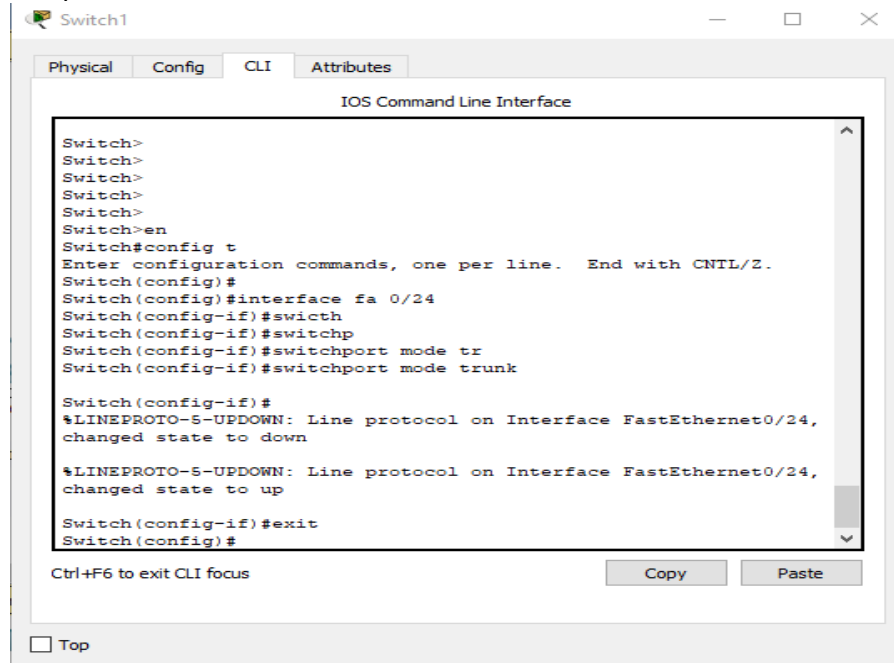


Figure 9: trunk mode enabling

5. After this provide IP addresses in all PC's as mentioned in the text, also provide subnet masks and default gateways. As provided in PC0, do same for all.

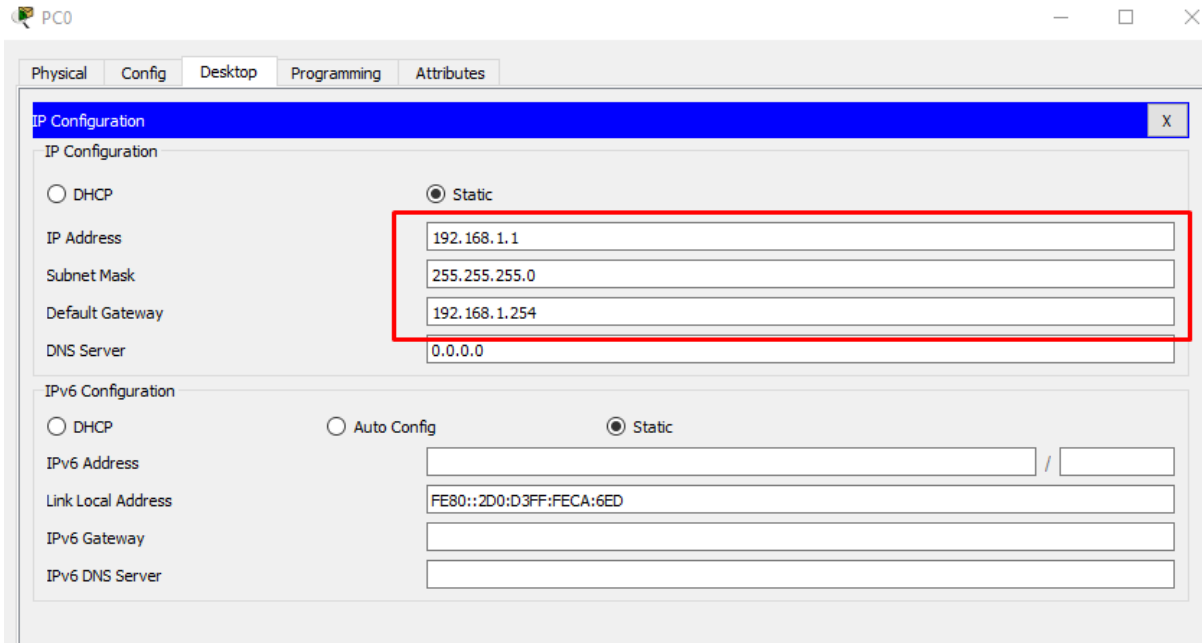


Figure 10: configuring computers

6. Now all the things necessary for the configuration of VLAN has been done, send PDU from PC0 in VLAN student to PC2 in VLAN student you will see that it will successfully sent, but if you sent from PC0 to PC1 or PC3 the PDU will be failed, hence this proved that you have configured VLAN's properly as the PC's or users cannot communicate cross VLAN's, but can communicate within same VLAN. Student can sac communicate with other students and faculty can communicate with other faculty but they cannot able to communicate with each other like student to faculty.

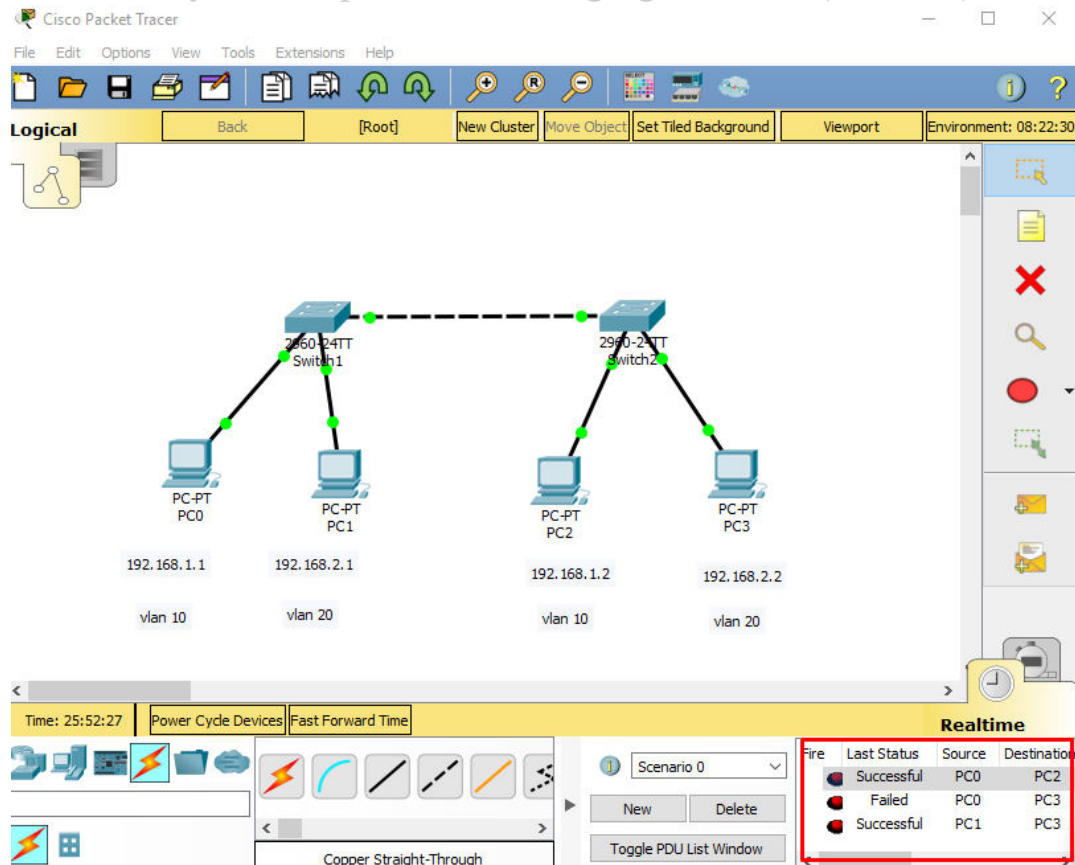


Figure 11: VLAN PDU result



National University of Computer and Emerging Sciences (NUCES) Islamabad

Access Control list

1. Introduction:

1.1. Standard Access Control List:

These are the ACLs in which filtering rules are defined using the source IP address only. These ACLs permit or deny the entire protocol suite of higher layers while filtering is applied on specific source IP address. Standard ACLs can't distinguish between the TCP, UDP and Https traffic of same source IP so all traffic of specific IP address will stay blocked. By using numbers 1-99 or 1300-1999 during configuration, router will understand it as a standard ACL and the specified address as source IP address.

1.2. Features:

- Standard Access-list is generally applied close to destination (but not always).
- In standard access-list, whole network or sub-network is denied.
- Standard access-list uses the range 1-99 and extended range 1300-1999.
- Standard access-list is implemented using source IP address only.
- If numbered with standard Access-list is used then remember rules can't be deleted. If one of the rule is deleted then the whole access-list will be deleted.
- If named with standard Access-list is used then you have the flexibility to delete a rule from access-list.

1.3. Types of Access Lists:

There are two ways in which Access control lists can be accomplished.

- Standard Access list
- Extended Access list

3. Walk through task:

Objective:

Construct a network topology with three routers, three switches, and six PCs. Configure Standard ACL to:

- Block the entire network behind Router 1 from accessing Router 3's network
- Block a specific PC from Router 2's network from accessing Router 3's network
- Allow Router 2's other PCs to access all networks normally

National University of Computer and Emerging Sciences (NUCES) Islamabad

Topology Setup:

To configure this, execute the following steps:

1. **Open Cisco Packet Tracer.**
2. **Add the following devices to the workspace:**
 - 3 Generic Routers (Router1, Router2, Router3)
 - 3 Generic Switches (Switch0, Switch1, Switch2)
 - 6 Generic PCs (2 PCs per switch)
3. **Make connections:**
 - Connect each pair of PCs to their respective switch using **copper straight-through wire**
 - Connect each router to its corresponding switch using **copper straight-through cable**
 - Connect routers together using **serial DCE/DTE cables**
4. **IP Addressing Scheme: Router 1 Network:**
 - Router1 Fa0/0: 192.168.10.1/24
 - PC0: 192.168.10.2/24 (Gateway: 192.168.10.1)
 - PC1: 192.168.10.3/24 (Gateway: 192.168.10.1)

Router 2 Network:

 - Router2 Fa0/0: 192.168.20.1/24
 - PC2: 192.168.20.2/24 (Gateway: 192.168.20.1)
 - PC3: 192.168.20.3/24 (Gateway: 192.168.20.1)

Router 3 Network (Destination/Protected Network):

 - Router3 Fa0/0: 192.168.30.1/24
 - PC4: 192.168.30.2/24 (Gateway: 192.168.30.1)
 - PC5: 192.168.30.3/24 (Gateway: 192.168.30.1)

WAN Links:

 - Router1 Se0/0/0 ↔ Router2 Se0/0/0: 10.0.0.1/30 ↔ 10.0.0.2/30
 - Router2 Se0/0/1 ↔ Router3 Se0/0/1: 10.0.1.1/30 ↔ 10.0.1.2/30
5. **Configure all router interfaces** with the IP addresses shown above.
6. **Configure RIP routing** on all routers.

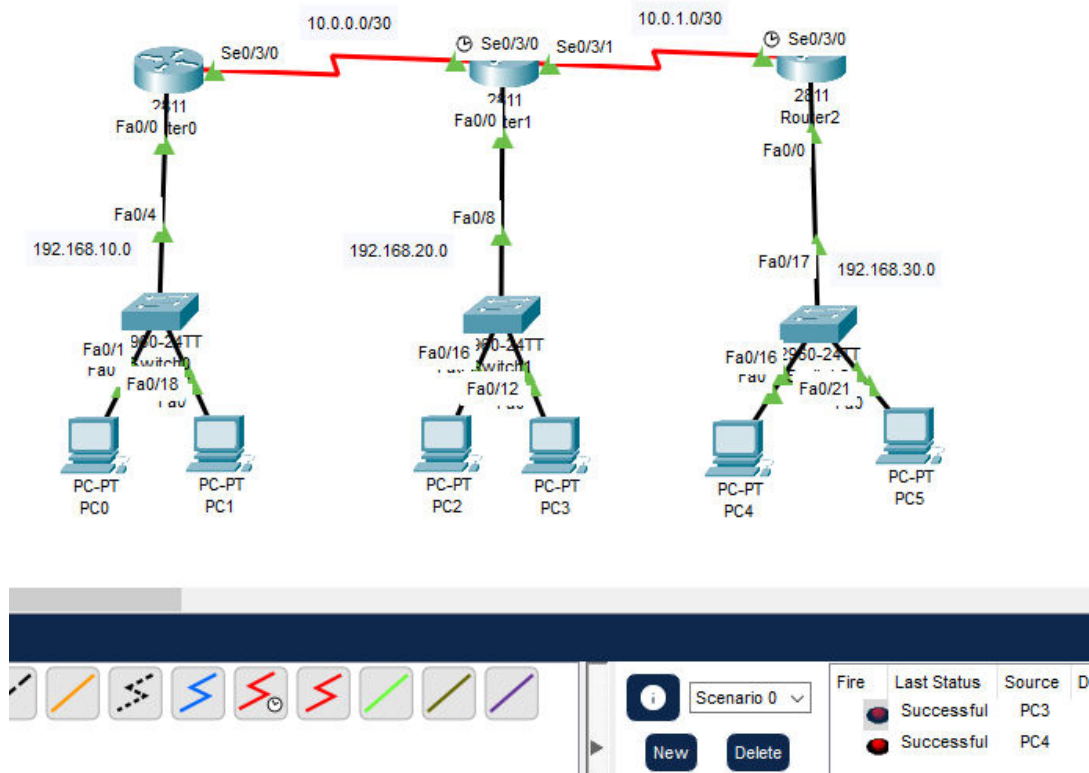


Figure 1: Routing.

Now after bringing the topology into working condition, we will now apply an ACL. As discussed earlier, two types of ACLs exist, but in this task we will only configure a **Standard ACL**.

Standard ACL filters traffic using only the source IP address and is usually applied near the destination.

Router 3's network (192.168.30.0/24) contains important servers. We want to:

- **Completely block** Router 1's entire network (192.168.10.0/24) from accessing Router 3's network
- **Block** only PC2 (192.168.20.2) from Router 2's network
- **Allow** PC3 (192.168.20.3) and all other traffic normally

Steps to Apply Standard ACL:

Step 1: Understanding Standard ACL Placement

Standard ACL uses source IP address only and should be applied close to the destination. Since we want to protect Router 3's network, we will apply the ACL on Router 3.

Step 2: Open Router 3 → CLI, go to global configuration mode:

```
Router> enable
Router# configure terminal
Router(config)#
```

Step 3.1: Block Single Host

Syntax: `access-list <ACL-number> deny host <source-IP>`

where (ACL number 1–99 = Standard ACL, 100–199 = Extended ACL)

Example:

Router3(config)# access-list 10 deny host 192.168.20.2

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 10 deny host 192.168.20.2
Router(config)#
```

Explanation:

- 10 = ACL number (1-99 for Standard ACL)
- deny = Block the traffic
- host 192.168.20.2 = Specific PC to block

Step 3.2: Block Entire Network

To block entire Router 1's network (192.168.10.0/24):

Syntax: `access-list <acl-number> deny <network-address> <wildcard-mask>`

Example:

Router3(config)# access-list 10 deny 192.168.10.0 0.0.0.255

```
Router(config)#
Router(config)#access-list 10 deny 192.168.10.0 0.0.0.255
Router(config)#
```

Step 3.3: Permit Remaining Traffic

After denying specific hosts/networks, we MUST permit all other traffic:

Syntax: `access-list <acl-number> permit any`

Example:

Router3(config)# access-list 10 permit any

```
Router(config)#  
Router(config)#access-list 10 permit any  
Router(config)#
```

Why is this needed?

- Every ACL has an implicit "deny all" at the end
- Without permit any, ALL traffic will be blocked!

Step 4: View the Access List

To verify your ACL configuration:

Syntax: `show access-lists`

Example:

```
Router(config)#  
Router(config)#  
Router(config)#exit  
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Router#show access-lists  
Standard IP access list 10  
  10 deny host 192.168.20.2 (2 match(es))  
  20 deny 192.168.10.0 0.0.0.255 (10 match(es))  
  30 permit any (38 match(es))  
  
Router#
```

Note: The ACL is created but NOT YET ACTIVE. ACLs don't work until applied to an interface.

Step 5: Apply ACL to Interface

For Standard ACL, apply close to destination.

Since traffic is coming from Router 2 towards Router 3, we apply the ACL on the **Serial interface** where traffic enters Router 3.

Syntax:

```
Router(config)# interface <interface-type> <interface-number>  
Router(config-if)# ip access-group <acl-number> <in|out>
```

Example:

```
interface Serial 0/3/0  
ip access-group 10 in
```

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial 0/3/0
Router(config-if)#ip access-group 10 in
Router(config-if)#exit
Router(config)#
```

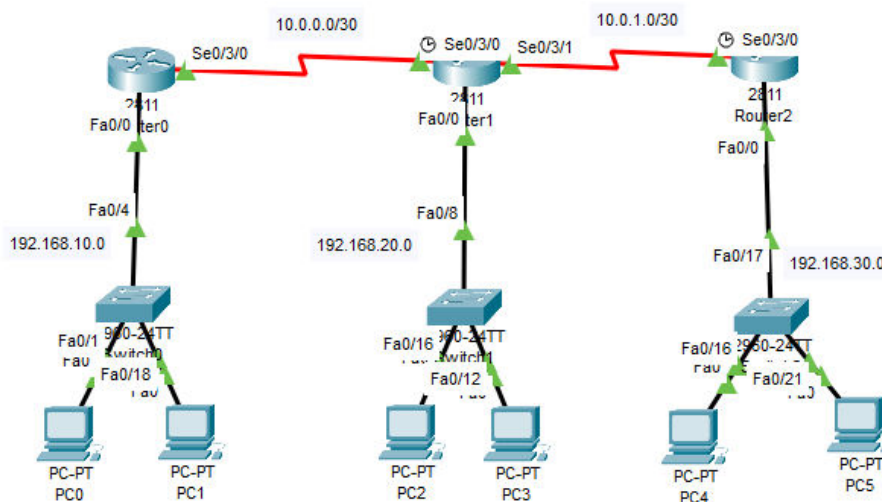
What does "in" mean?

- in = Filter packets entering the router from this interface
- out = Filter packets leaving the router from this interface

In our case:

- Traffic from Router 2 enters Router 3 via Serial 0/3/0
- We want to filter it before it reaches Router 3's LAN
- So we use "in" direction

Now test using PDU or the ping command to verify whether Router 3 is blocking the R1 network traffic and one PC of R2 correctly.



Fire	Last Status	Source	Destination	Type	Color
Failed	Failed	PC0	PC4	ICMP	Red
Failed	Failed	PC2	PC4	ICMP	Red
Successful	Successful	PC3	PC5	ICMP	Green